

Quiz 1: Agents and Search

Weeks 1-2

CMPSC 480 - Student Question Deck

Assessment overview

Demonstrate concise reasoning about agent models and classical search guarantees.

Coverage: Weeks 1-2

Points

20 total

Expected time

30 minutes

Submission

Individual PDF

Learning objectives

- ▶ Apply PEAS and search formulation.
- ▶ Trace frontier order.
- ▶ Compare guarantees.
- ▶ Reason about admissible heuristics.

Requirements checklist

- ▶ Work independently.
- ▶ Show intermediate frontier or arithmetic steps.
- ▶ State every assumption used for a guarantee.
- ▶ No electronic communication.

Task 1 - Agent formulation (4 points)

Analyze a warehouse-routing agent.

a. Give one element for each PEAS component.

[2 points]

b. Explain why rational does not mean omniscient.

[2 points]

Task 2 - BFS and DFS trace (4 points)

Successors are generated alphabetically. S connects to A and B; A to G; B to C; C to G.

- a. Trace BFS until a goal is removed. [2 points]
- b. Trace DFS when successors are pushed in alphabetical order. [2 points]

Task 3 - Cost-sensitive search (6 points)

Two routes reach a goal: S-A-G costs $5+1$, and S-B-C-G costs $1+1+1$.

- a. Which route can BFS return, and why is that not a minimum-cost guarantee? [2 points]
- b. Trace UCS priorities until the goal is removed. [2 points]
- c. State conditions under which UCS is complete and optimal. [2 points]

Task 4 - A-star heuristic (6 points)

A four-neighbor grid uses unit costs and Manhattan distance.

- a. Define f , g , and h in A-star. [2 points]
- b. Prove Manhattan distance is consistent. [2 points]
- c. If two nodes have $(g,h)=(4,3)$ and $(5,1)$, which is removed first? [2 points]

Edge cases and defensive programming

- ▶ A start state is already a goal.
- ▶ A graph contains a cycle.
- ▶ Two A-star nodes have equal f .

Submission instructions

- ▶ Write your name and assumptions.
- ▶ Show work for traces and arithmetic.
- ▶ Submit only the question-response PDF.

Do not submit credentials, generated caches, or unapproved libraries.

Grading rubric

Category	Points	Required evidence
Agent formulation	4	Written responses.
BFS and DFS trace	4	Frontier trace and returned paths.
Cost-sensitive search	6	Algorithm choice and cost trace.
A-star heuristic	6	Proof sketch and comparison.

Reflection prompt

In one sentence:

Name the assumption most often omitted when stating a search guarantee.